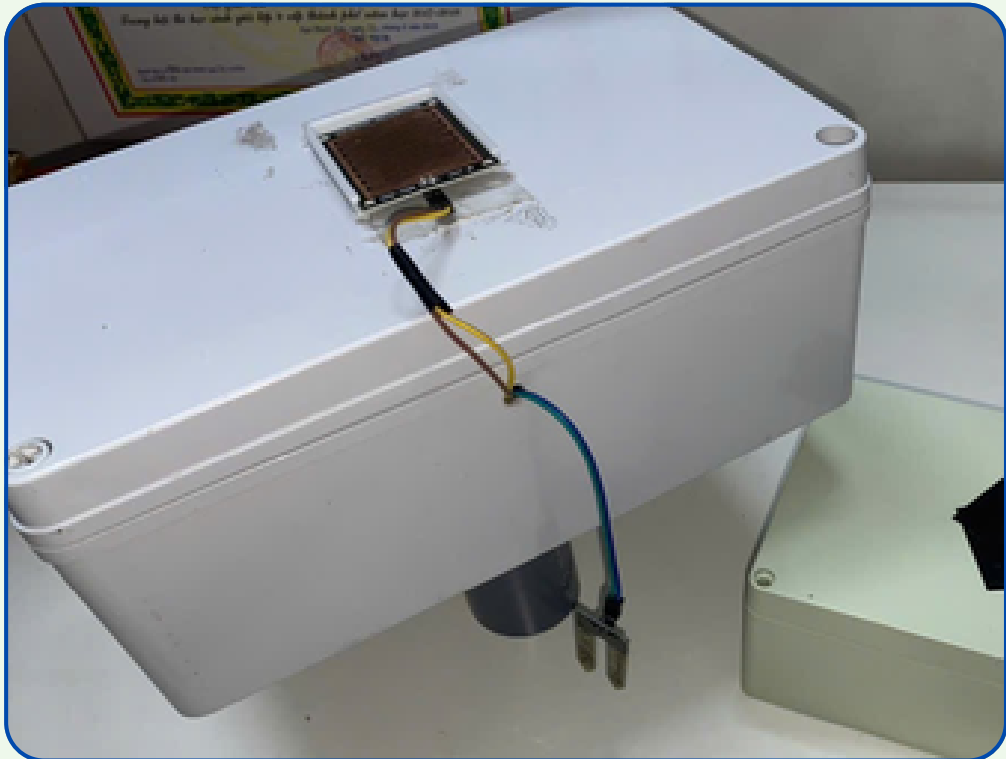


WORKING PRINCIPLE

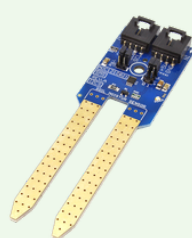
SENSOR DEVICE



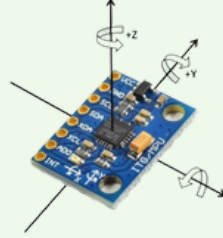
- Collect data and calculate landslide risk levels.
- Installed in areas with high landslide risk, steep terrain or frequent heavy rains.
- Operated by backup batteries, can be upgraded to solar power in the future for increased sustainability..



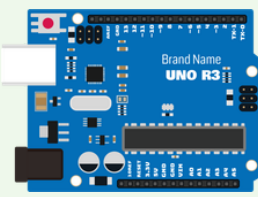
Rain sensor



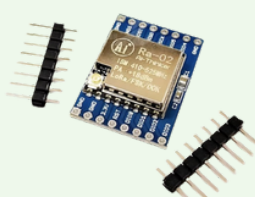
Moisture Sensor



Accelerometer Sensor



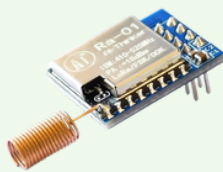
Arduino Uno R3



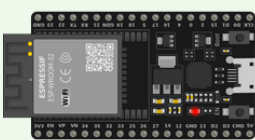
LoRa Transmitter Module

WARNING DEVICE

- Receive data from sensor device via LoRa transmission line to LoRA receiver module
- Located in residential areas, environmental exploitation enterprises with high risk of landslides
- When the warning level is $\geq 70\%$, the system will activate the warning speaker, helping people to evacuate and respond in time.



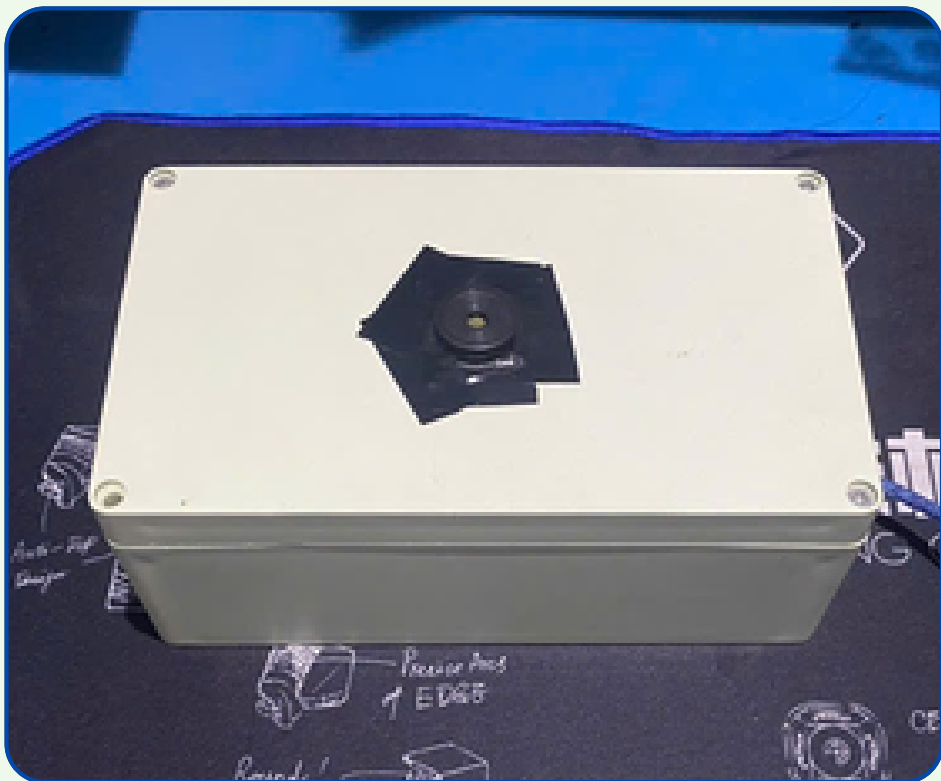
Lora receiver module



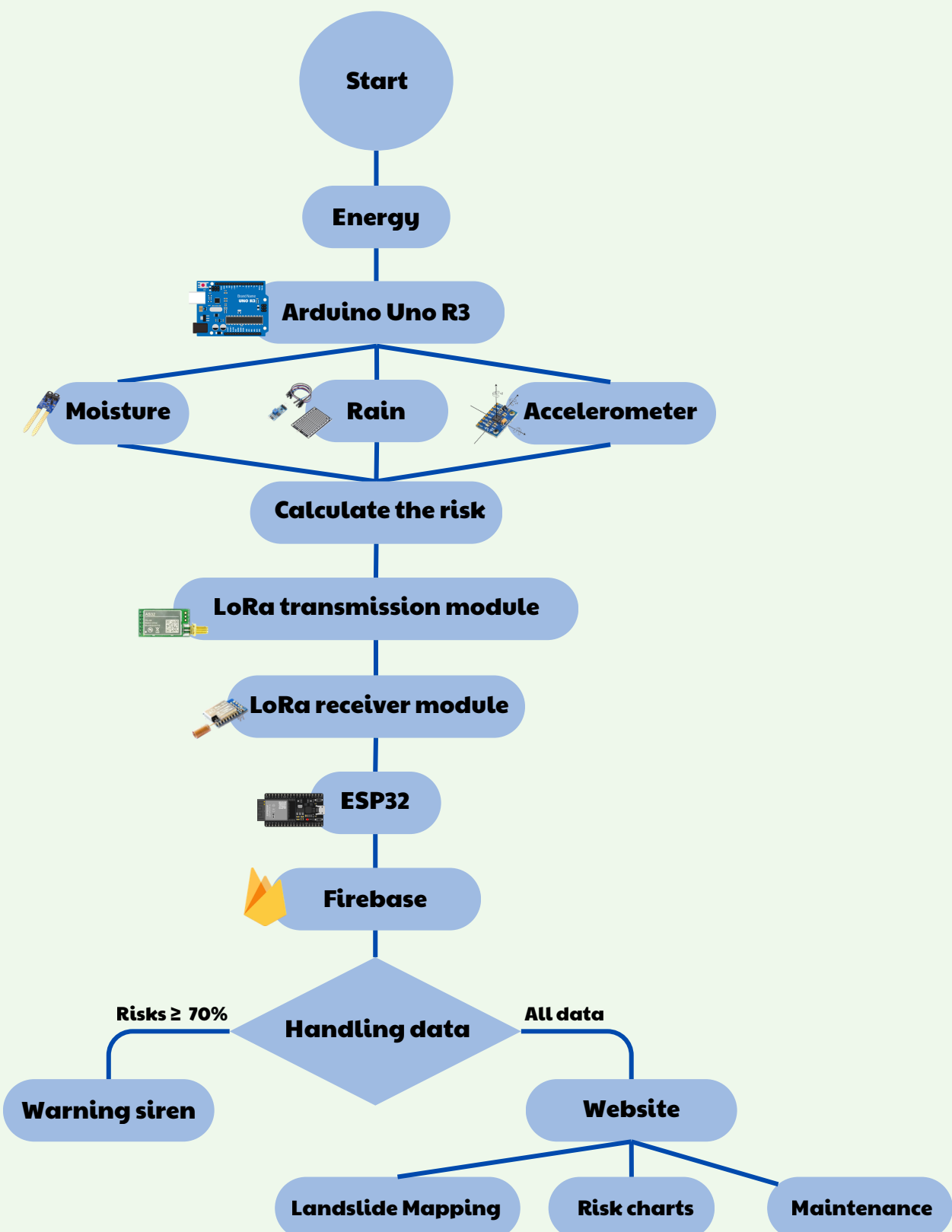
ESP32



Warning siren



OPERATION DIAGRAM



FORMULA FOR CALCULATING RISK

$$\text{Risk} = \alpha \cdot R_c + \beta \cdot M + \gamma \cdot S + \delta \cdot \theta + \varepsilon \cdot H$$

Factor	Explanation
Accumulated rainfall (Rc)	Lượng mưa lớn trong thời gian ngắn hoặc kéo dài làm tăng áp lực nước trong đất.
Rainfall duration (θ)	Mưa kéo dài khiến đất bị bão hòa, giảm khả năng giữ kết cấu.
Soil moisture (M)	Độ ẩm cao khiến đất mất kết dính, dễ trượt.
Terrain slope (S)	Địa hình dốc làm gia tăng trọng lực kéo đất xuống.
Seismic activity (H)	Động đất gây rung chấn, phá vỡ ổn định đất.